Name Key

Physics 390 In-Class Activity 8

1. Fill in the blanks with the technique that best fits the description. Your options are:

Brightfield, Phase Contrast, Darkfield, Polarization, Wide-field Fluorescence, LSCM, SDCM (spinning disk), and Deconvolution

(A) Generates deblurred images using a single pinhole LSCM

(B) Is used to visualize amplitude specimens, such as fixed, stained cells BF

(C) Produces contrast by shifting the zeroth order (undiffracted) wave by 90° PC

(D) Generates fluorescence images of thick samples that are plagued by blur WFF

2. Fill in the blanks with the technique that best fits the description. Your options are:

Two-photon, STED, SIM, and PALM

(A) Is a premier method of deep imaging 2p

(B) Achieves super-resolution using stochastic switching PALM

(C) Surpasses the diffraction limit by a factor of two in all three dimensions SIM

(D) Achieves super-resolution by quenching fluorescence using stimulated emission STED

3. Which of the following are components of Abbe theory (Circle all correct answers)?

(A) If only undiffracted light is collected by the objective, the image is unresolved

(B) The image can be altered by spatial filtering in the intermediate image plane

(C) Spatial resolution is given by the formula NA/λ

(D) Diffraction orders appear in the back focal plane

4. Which of the following are true of two-photon microscopy (Circle all correct answers)?

(A) Detector is a camera ✗

(B) Does not require a pinhole

(C) Excitation is achieved using ultraviolet photons

(D) Fluorescence is confined to laser focus

5. Which of the following statements are true (Circle all correct answers)?

(A) In Darkfield microscopy, the oblique illumination misses the objective, and the specimen diffracted light enters the objective

(B) To achieve an axial resolution of 500 nm, you should sample every 500 nm

(C) In Köhler's method, uniform illumination is created by focusing the filament image onto the FFP of the condenser

(D) A finely spaced grating will generate closely spaced diffraction spots

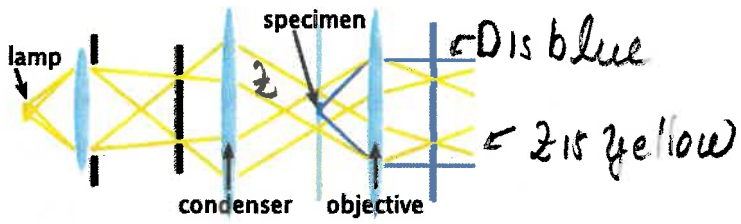
6. A schematic of a phase contrast microscope is shown below. Clearly identify:

(A) The zeroth order rays by superimposing the word (Z) on these rays and color

(B) The diffracted rays by superimposing the word (D) on these rays and color

(C) The location where rays are phase shifted by the phase plate by circling the location and writing "phase shift"

(D) The location where illumination is made annular by circling the location and writing "annular"



7a. List the specimen-containing conjugate planes in the Köhler pathway.

FO, specimen, int. image, retina

7b. List the light source-containing conjugate planes in the Köhler pathway.

light source, condenser FFP, objective BFP, eye pupil

8. Which are true of the techniques STED and PALM (Circle all correct answers)?

(A) Switching molecules between bright and dark states plays a key role in surpassing the diffraction limit

(B) STED involves no post-acquisition image processing

(C) PALM is well-suited to imaging fast dynamic processes

(D) Their inventors were awarded the Nobel Prize

9. Which are true of a compound microscope (Circle all correct answers)?

(A) The specimen is positioned less than $f_{objective}$ from the objective

(B) The objective produces a magnified real, inverted image

(C) There is no limit on useful magnification

(D) The eyepiece produces a magnified, virtual image

10. Which are true of fluorescence microscopy (Circle all correct answers)?

(A) Generally requires labeling of cellular constituents

(B) Lost rotational energy gives rise to the Stoke's shift

(C) A transmitted light path is preferred

(D) Is the dominant optical microscopy imaging mode

(E) Is well-suited to multi-color imaging

11. Which are true of diffraction-limited resolution (Circle all correct answers)?

(A) Resolution improves with decreasing wavelength

(B) Resolution improves with decreasing numerical aperture

(C) Objects with dimensions smaller than a diffraction-limited spot (the Airy disk) appear to be larger than their true size

(D) It is impossible to surpass diffraction-limited resolution